

Void Galaxies and Their Properties

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Introduction

The large-scale structure of the universe is not uniform. Galaxies are distributed in a complex cosmic web composed of dense clusters, filaments, thin walls, and vast under dense regions known as cosmic voids. Understanding how the local environment shapes a galaxy's properties, its color, morphology, star formation rate, and chemical composition, is one of the central questions of modern galaxy formation theory.

Galaxies in dense environments, such as clusters and filaments, are subject to a variety of interactions; tidal forces, dynamical friction, and mergers with neighboring galaxies. These processes complicate efforts to isolate the baseline physics of galaxy formation. Void galaxies, by contrast, exist in extreme under-dense regions and rarely experience such interactions. For this reason, they have been described as 'fossils' of early galaxy formation, retaining properties more representative of primordial conditions than galaxies in denser environments (2; 4).

Observational studies using the Sloan Digital Sky Survey (SDSS) have established that void galaxies tend to be bluer, of later morphological type, and more actively star-forming than comparable galaxies in walls (1; 4). More recently, hydrodynamical simulations have been used to model galaxy formation directly within the void environment, revealing that galactic outflows, large-scale winds of gas driven by supernova feedback, are significantly less prevalent in voids than in field environments (3). Meanwhile, work on void-finding methodology has demonstrated that conclusions about void galaxy properties depend sensitively on the definition of a void in the first place (5).

Cosmic Voids

A cosmic void is an expansive region of space that contains far less matter than the universe's average. Conceptually, a void can be thought of as a hole or bubble within the cosmic web,

the vast filamentary network of matter that spans the observable universe. This web is composed of three primary structural components: dense, compact knots (galaxy clusters) at the intersections of filaments; elongated filaments connecting those knots; and flat, sheet-like walls separating the enclosed void regions. The voids themselves are the spaces enclosed by this network.

Voids are not perfectly empty. Their interiors contain diffuse gas clouds, dark matter, rogue stars, and, galaxies. The density contrast within a void is typically characterized by the parameter

$$\frac{\delta\rho}{\rho} < -0.6, \quad (1)$$

where $\delta\rho = \rho_{\text{local}} - \bar{\rho}$ is the difference between the local matter density ρ_{local} and the cosmic mean density $\bar{\rho}$. A value of $\delta\rho/\rho < -0.6$ therefore means the local density is at least 60% below the cosmic mean. At the centers of the deepest voids this contrast can reach -0.9 (1). Despite this under-density, voids are not structurally featureless; they contain internal density variations, small-scale filamentary connections between void galaxies, and nested sub-voids within larger parent voids.

Voids are enormous. In the SDSS Data Release 7, Hoyle et. al. (1) analyzed a sample derived from 1,054 statistically significant voids with radii larger than $10 h^{-1}$ Mpc, with the largest exceeding $30 h^{-1}$ Mpc in effective radius, where $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$ is the dimensionless Hubble parameter. Collectively, voids occupy approximately 62% of the survey volume, dominating the universe by volume even as they account for only a small fraction of its total matter.

Identifying voids from galaxy survey data requires a void-finding algorithm. Two widely used algorithms are VoidFinder, which grows and merges overlapping spherical void regions, and V² (Voronoi Voids), a Python implementation of the ZOBOV (ZOnes Bordering On Voidness) algorithm that makes no assumptions about void shape. VoidFinder and V² can classify the same set of galaxies very differently, with direct consequences for conclusions about void galaxy properties (5)

Void Galaxies

Definition and Formation Environment

A void galaxy is a galaxy located within the interior of a cosmic void, residing in a region satisfying Equation (1) (4). Because the void environment lacks the dense streams of matter found in filaments and walls, void galaxies grow primarily through accretion, the gradual in-fall of gas from the surrounding intergalactic medium, rather than through mergers with other galaxies. Accretion is the process by which in-falling gas loses angular momentum and settles onto a growing galaxy, fueling star formation without the violent morphological disruption associated with mergers. Merger events, which drive rapid morphological transformation and can trigger bursts of star formation or active galactic nucleus (AGN) activity, which are rare or absent in void environments.(3)

Void galaxies are instead fed by low-velocity gas filaments that supply material for gradual galactic growth. The internal density variations within voids can produce gravitationally bound dark matter halos that eventually host galaxies. Although voids are globally under dense, these local over densities can themselves develop sub-void structure on smaller scales. Despite their isolated origins, void galaxies display the same overall structural organization as galaxies in denser environments, disk morphologies, central bulges, and spiral arms, reflecting the universality of gravitational collapse as a formation mechanism (3).

Photometric Properties

The photometric properties of void galaxies have been studied extensively using SDSS imaging data. Hoyle et. al. (1) analyzed the colors and morphologies of 88,794 void galaxies identified in SDSS DR7 and compared them to a matched sample of wall galaxies with the same luminosity distribution. Using the $u - r$ color as a proxy for star formation activity, since young, hot, massive stars radiate predominantly in the ultraviolet and blue optical, the authors found that void galaxies are statistically bluer at every luminosity bin from bright to dwarf galaxies and that voids contain more spirals (see Figure 1).

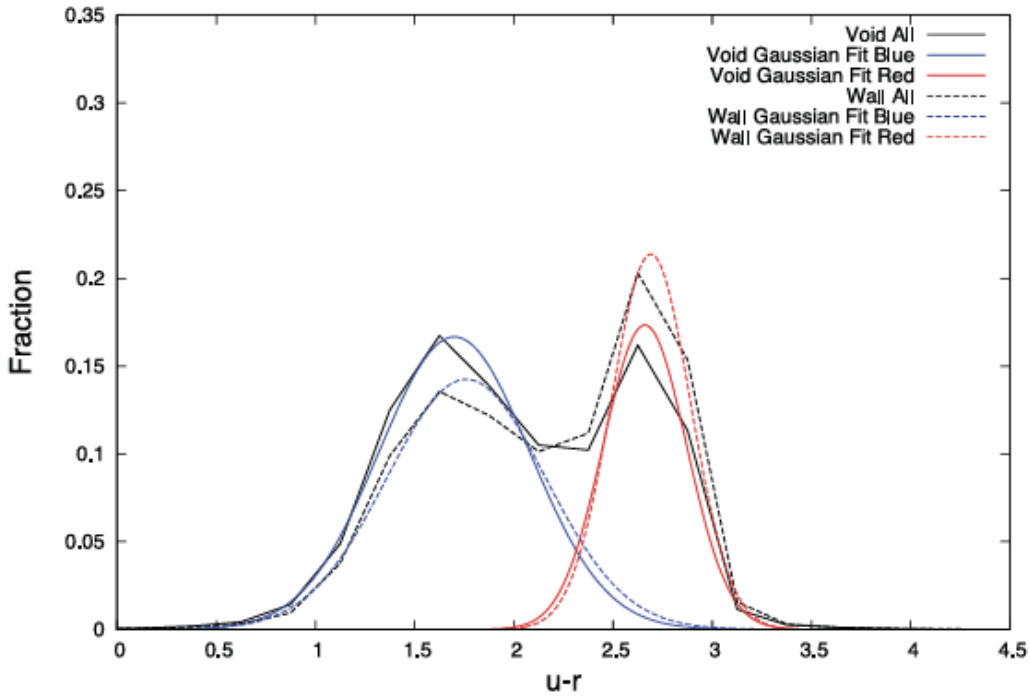


Figure 1: Distribution of $u - r$ colors for void galaxies (solid lines) compared to wall galaxies (dotted lines). This diagram shows the split between blue, spiral type galaxies and red, elliptical galaxies. There are more elliptical galaxies in the walls and more spirals in the voids. It also shows the tendency for void galaxies to be bluer compared to wall galaxies. Adapted from Hoyle et. al. (1).

The fraction of blue galaxies increases from bright to dwarf galaxies in both voids and walls, consistent with the general color-mass trend in the galaxy population. However, at

every luminosity, the void population contains a higher fraction of blue galaxies than the wall population. Late-type void galaxies, disk-dominated systems, were found to be bluer and of lower concentration than their wall analogs, indicating that the void environment preferentially hosts morphologically late, star-forming systems and suppresses the development of early-type (elliptical) morphologies common in denser environments.

The choice of void-finding algorithm significantly affects these conclusions. Zaidouni et. al. (5) showed that galaxies classified as void galaxies by VoidFinder are reliably bluer and fainter than wall galaxies, consistent with the photometric results above. However, galaxies classified as void galaxies by the V^2 algorithm showed no statistically significant photometric differences compared to wall galaxies. This discrepancy arises because V^2 voids tend to extend their boundaries into the dense walls between voids, inadvertently including wall galaxies in the void sample and diluting any void-specific signal.

Spectroscopic Properties and Star Formation

Rojas et. al. (4) conducted a spectroscopic analysis of 103 void galaxies identified in the SDSS and compared their emission line properties to a sample of wall galaxies of similar luminosity. Emission line transitions such as $H\alpha$ (hydrogen recombination), $[O II]$, $H\beta$, and $[O III]$ trace ionized gas in star-forming $H II$ regions; their equivalent widths therefore serve as indicators of current star formation activity. Void galaxies were found to have systematically larger equivalent widths in these lines than wall galaxies, indicating higher specific star formation rates (SSFRs), more star formation per unit stellar mass.

A complementary diagnostic, the 4000-Å break strength $D_n(4000)$, measures the depth of the spectral discontinuity produced by metal-line absorption in the atmospheres of older, cooler stars. A small $D_n(4000)$ value indicates a younger, hotter stellar population. Void galaxies exhibit smaller $D_n(4000)$ values than wall galaxies, confirming that they host younger stellar populations on average. This is consistent with the picture of void galaxies as more primordial systems that have undergone less cumulative star formation and thus retain a higher fraction of recently formed stars.

Moorman et. al. (2) extended this analysis using a larger sample from SDSS DR8, supplemented by ultraviolet photometry from the Galaxy Evolution Explorer (GALEX) to measure star formation on timescales of ~ 100 Myr. Their results confirmed higher SSFRs in void galaxies at fixed stellar mass, particularly for low-mass systems. They also found that void galaxies tend to be more gas-rich and have lower stellar mass surface densities, consistent with a formation history that has proceeded more slowly and with less external stimulation than in wall environments.

Void galaxies also tend to have lower metallicities than wall galaxies of the same mass. Metallicity, the abundance of elements heavier than hydrogen and helium, often denoted Z , increases over time as successive generations of stars synthesize heavy elements in their cores and return that material to the interstellar medium through stellar winds and supernova explosions. The lower metallicities of void galaxies thus reflect their younger effective ages and the smaller number of stellar generations that have enriched their gas reservoirs.

AGN activity, the energetic output produced when a supermassive black hole actively accretes material at a galactic center, is also less common in void galaxies than in wall galaxies of comparable mass. In dense environments, galaxy-galaxy interactions and mergers

can funnel gas toward galactic centers and trigger AGN episodes. In voids, where such interactions are rare, this fueling mechanism is largely absent.

Simulated Properties and Galactic Outflows

Pan et. al (3) presented a dedicated hydrodynamical simulation of galaxy formation within a cosmic void, using the RAMSES adaptive mesh refinement code. RAMSES solves the equations of hydrodynamics and gravity on a dynamically refined Cartesian mesh, allowing high spatial resolution in regions of interest while maintaining a large simulation volume. Their simulation modeled a void of diameter $120 \text{ cMpc } h^{-1}$ with a density contrast of $\delta\rho/\rho \approx -0.8$, identified as the lowest-density region among 1,000 random spheres drawn from a 1 Gpc parent simulation. A companion field simulation of the same box size was run for comparison.

One of the most striking results concerned galactic outflows. Galactic outflows are large-scale winds of gas expelled from galaxies by supernova feedback, the kinetic and thermal energy deposited into the interstellar medium by core-collapse supernovae, and by stellar radiation pressure. These winds can enrich the circum-galactic and intergalactic medium with metals and regulate star formation by removing gas from the galaxy. Pan et. al (3) developed an analytical framework for computing the wind volume filling fraction f_{wind} , defined as the fraction of the total intergalactic volume swept by outflow material. In the field simulation, $f_{\text{wind}} \approx 18.6\%$ at infinite resolution, while in the void simulation this fell to just $f_{\text{wind}} \approx 3.1\%$, reflecting the lower number density and lower masses of galaxies in void environments, which drive correspondingly weaker winds.

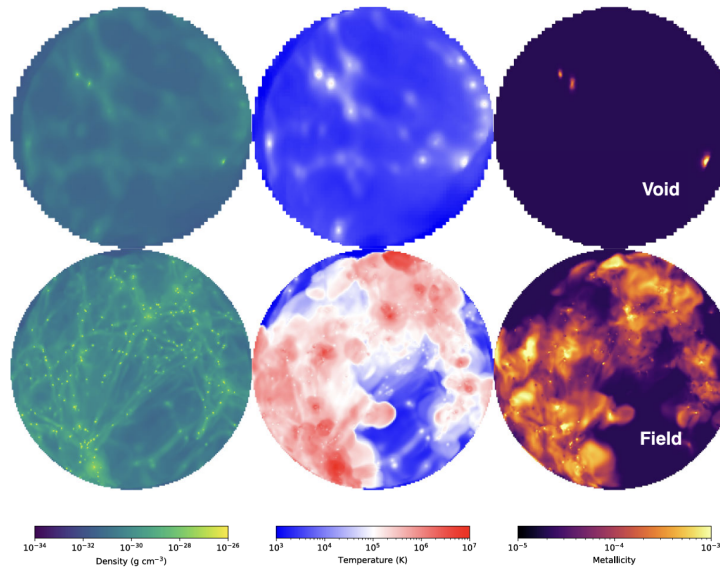


Figure 2: Cosmic void environment (top) compared to a field environment (bottom) of the gas density (left), gas temperature, (middle), and gas metallicity (right). For a $25 \text{ cMpc } h_0^{-1}$ sphere in the void and field, both projected along the x-axis. Adapted from (3).

This result has direct implications for the chemical evolution of the intergalactic medium

within voids. Because void galaxies produce weaker outflows and experience fewer mergers, the surrounding gas retains more of its primordial, low-metallicity composition, consistent with observations of lower gas-phase metallicities in void galaxy samples (see Figure 2).

Importance to Cosmology

Cosmic voids and their galaxies are of interest to cosmology for several interrelated reasons. First, because void galaxies evolve with minimal interference from mergers and environmental processing, they offer a more direct window onto the intrinsic physics of galaxy formation. They may serve as a baseline for understanding how galaxies would form in the absence of environmental effects, a kind of control group against which the impact of denser environments can be quantified (2).

Second, the statistical properties of voids, their size distribution, radial density profiles, and internal galaxy populations, are sensitive probes of the underlying cosmological model. Measurements of void properties constrain the growth rate of large-scale structure, the equation of state of dark energy, and models of modified gravity. Upcoming surveys such as Euclid will produce void catalogs of unprecedented size and depth, making the study of void galaxies increasingly central to precision cosmology (3).

Third, the methodological challenges highlighted by Zaidouni et. al. (5) underscore the importance of standardizing void-finding procedures across the field. If different algorithms yield qualitatively different conclusions about void galaxy properties, then apparent discrepancies between observational studies may be artifacts of methodology rather than genuine physical differences. Resolving this issue will require both improved theoretical understanding of void boundaries and community consensus on best practices for void identification.

Taken together, the emerging picture is one of void galaxies as genuinely distinct systems: younger, bluer, more gas-rich, and less chemically enriched than their counterparts in denser regions. As simulations improve in resolution and surveys expand in depth, the void environment will continue to provide one of the clearest available views into the earliest stages of galaxy formation.

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